



Individual Work

Lab. No: 10 , Assignment No.: 1

Topic: SVM/KNN/DT

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Program No: 10

Program Title:

For Implementing DT, SVM etc. You have to mention IN THE REPORT

(For each ML Algorithms do the following)

a) Problem Statement

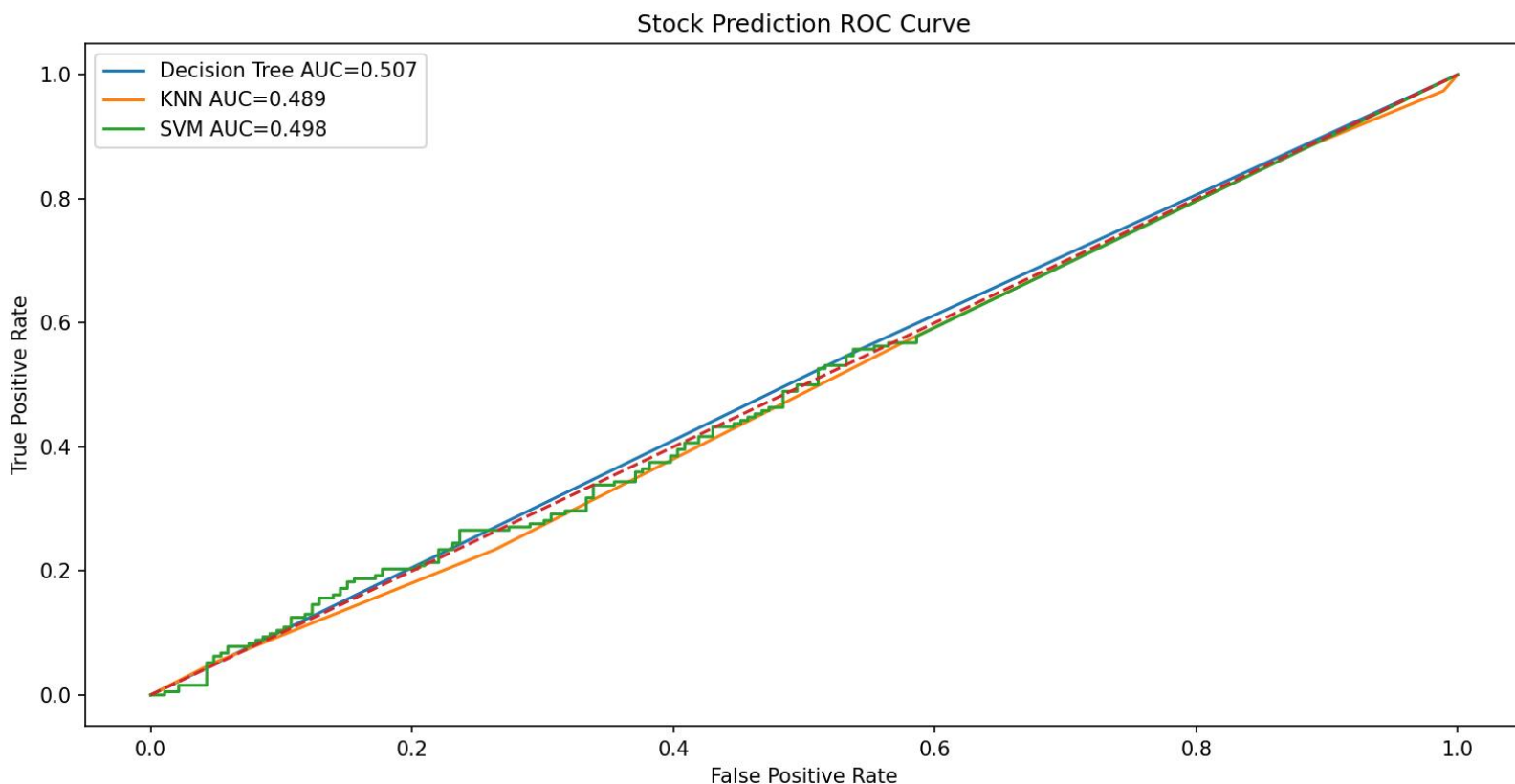
b) Dataset Used (Link for the repository)

c) ML Algorithm Used: DT, KNN, SVM etc.

d) Result & Discuss (Here you have to show AUC, ROC curve, and comparison table & graph related to Performance measure like Accuracy, Specifiicity, Sensitivity, F1-Score etc.)

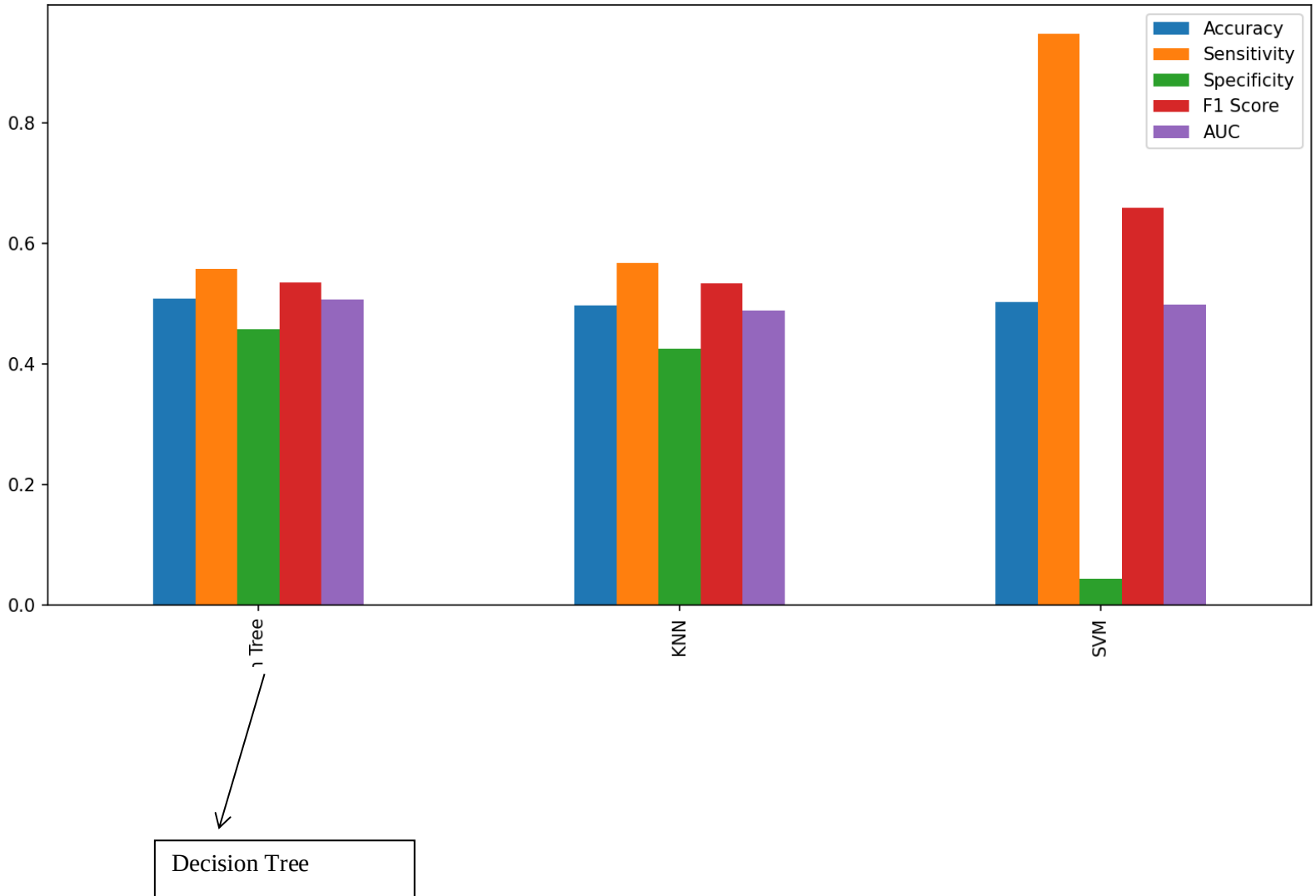
Input/Output Screenshots:

RUN-1:



RUN-2

Stock Market ML Comparison

**Results:****Performance Comparison**

	Model	Accuracy	Sensitivity	Specificity	F1 Score	AUC
0	Decision Tree	0.507937	0.557292	0.456989	0.535000	0.507140
1	KNN	0.497354	0.567708	0.424731	0.534314	0.489261
2	SVM	0.502646	0.947917	0.043011	0.659420	0.497830

Source code

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import roc_curve, auc
from sklearn.metrics import f1_score
df = pd.read_csv("all_stocks_5yr.csv")
print(df.head())
apple = df[df['Name']=='AAPL']
apple['Tomorrow_Close'] = apple['close'].shift(-1)
apple['Target'] = (
apple['Tomorrow_Close'] > apple['close']
).astype(int)
apple.dropna(inplace=True)
X = apple[[
'open',
'high',
'low',
'close',
'volume'
]]
y = apple['Target']
X_train,X_test,y_train,y_test = train_test_split(
X,
y,
test_size=0.3,
random_state=42
)
scaler=StandardScaler()
X_train=scaler.fit_transform(X_train)
X_test=scaler.transform(X_test)
models={
"Decision Tree":
DecisionTreeClassifier(),
"KNN":
KNeighborsClassifier(n_neighbors=5),
"SVM":
SVC(probability=True)
}
results=[]
plt.figure(figsize=(10,7))

for name,model in models.items():

```

```

model.fit(X_train,y_train)
pred=model.predict(X_test)
prob=model.predict_proba(X_test)[: ,1]
accuracy=accuracy_score(
y_test,
pred
)
f1=f1_score(
y_test,
pred
)
tn,fp,fn,tp = confusion_matrix(
y_test,
pred
).ravel()
sensitivity = tp/(tp+fn)
specificity = tn/(tn+fp)
fpr,tpr,_=roc_curve(
y_test,
prob
)
roc_auc = auc(
fpr,
tpr
)
results.append([
name,
accuracy,
sensitivity,
specificity,
f1,
roc_auc
])
plt.plot(
fpr,
tpr,
label=name+" AUC="+str(round(roc_auc,3))
)
plt.plot([0,1],[0,1], '--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("Stock Prediction ROC Curve")
plt.legend()
plt.show()
columns=[
"Model",
"Accuracy",
"Sensitivity",
"Specificity",
"F1 Score",
"AUC"

```

```
]
df_result = pd.DataFrame(
    results,
    columns=columns
)
print("\nPerformance Comparison\n")
print(df_result)
df_result.set_index("Model").plot(
    kind="bar",
    figsize=(10,6)
)
plt.title("Stock Market ML Comparison")
plt.show()
```